

Pairwise Independence of Internal Splitting, Dichotomy, and Composition in Finite Extensional Magmas

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Abstract

Internal splitting (S), the classifier dichotomy (D), and internal composition (C) are three natural algebraic properties of finite extensional 2-pointed magmas. We prove they are pairwise independent: no one implies any other. Lean-verified finite counterexamples at sizes 4 through 10 establish all six non-implications, four with provably tight bounds. A joint irredundance result strengthens this: all eight Boolean combinations of (S, D, C) are realized, so no property is a Boolean function of the other two. The minimum coexistence witness has $N=5$ and is optimal, since the ICP requires 3 pairwise distinct core elements.

The setting is fixed by four universal obstructions: nontrivial combinatory algebras with s and k must be infinite; associativity is incompatible with combining a classifier and a retraction pair; commutativity is incompatible with two distinct left-absorbers; and a single left-absorber collapses any classifier. What survives is finite, non-associative, non-commutative, extensional 2-pointed magmas.

Further structural results: the three-category decomposition induced by D is an isomorphism invariant; the automorphism group of any $N=5$ S+D+C magma has order at most 2, with the unique possible non-trivial element being the classifier transposition $(\tau_1 \ \tau_2)$; a size-driven phase transition takes the role-shape from a single forced configuration at $N=5$ (every retraction pair satisfies $s = r$, every C-triple has the form (τ_1, g, τ_2)) to a heterogeneous landscape at $N \geq 6$ where strong S re-opens and multiple shape classes coexist; and the ICP is logically equivalent to the standard Compose+Inert axioms. Lean-verified results — independence counterexamples, joint-irredundance witnesses, decomposition invariance, the ICP equivalence, the all- N scaling witness, H-triple type-coherence, the $N=5$ structure theorem, and the mirror-row theorem — compile with zero sorry; the remaining universal- N algebraic results in Section 3 not yet transcribed to Lean have hand proofs supported by Z3 UNSAT certificates.

Keywords

finite extensional magma, pairwise independence, classifier dichotomy, internal composition, Lean 4, machine-checked proofs

1 Introduction

A finite magma is the Cayley table of a binary operation, stripped of syntax and types. We study a particular kind: *finite extensional algebras with binary judgment* — finite magmas whose binary operation returns, alongside data outputs, judgments drawn from a fixed two-element subset of the carrier playing the role of truth values.

Preprint version. The Lean proofs, SAT search scripts, and manuscript text were developed with assistance from Claude (Anthropic). All results have been independently verified: Lean proofs compile with `lake build` (zero sorry), and SAT counterexamples are frozen in the repository for reproducibility. The author takes full responsibility for all contents.

Within this setting we ask three natural operational questions: can the algebra *represent* its own data, can it *classify* its own structure, and can it *compose* its own operations. Each question has a precise formal answer (S, D, C), and our main result is that the three are pairwise independent: no one implies any other.

The setting is fixed by three structural commitments, all of them positive. *Extensionality* is the foundational one: an element’s identity is exhausted by its row in the Cayley table, so an element *is* its action under the operation [6]. *Two distinguished truth values* z_1, z_2 are the minimum number for the operation to register a binary outcome; equipping the algebra with judgment cardinality one is degenerate (Remark 1), and higher arities are natural generalisations we do not pursue. *Inertness of the truth values* formalises what makes them truth values rather than data: $z_i \cdot x = z_i$ for all x , so post-judgment behaviour depends only on judgment status. The truth values are operationally distinguishable (they absorb), structurally interchangeable (no axiom prefers z_1 to z_2), and exhaustive among inert elements (no third element is left-absorbing). Together these commitments select the class of finite extensional 2-pointed magmas.

The three capabilities. Let $\text{core}(S) = S \setminus \{z_1, z_2\}$ denote the carrier minus its truth values. *Internal splitting* (S): there exists a section/retraction pair $(s, r) \in \text{core}(S)^2$ with $r \cdot (s \cdot x) = x$ on $\text{core}(S)$, so the algebra has internal encoding/decoding for its own data. *Internal dichotomy* (D): every element of core has the same judgment behaviour across all of core — either every output lies in $\{z_1, z_2\}$ (a *classifier*) or no output does (a *non-classifier*) — and at least one example of each type exists. *Internal composition* (C): there exist three pairwise distinct core elements a, b, c with b core-preserving and $a \cdot x = c \cdot (b \cdot x)$ on $\text{core}(S)$, with a taking at least two distinct values — a non-trivial factorization in the left-regular representation. Each capability is what an external machine would otherwise have to supply (naming and decoding, binary judgment, operation chaining), realised instead as a behaviour of elements of the algebra modulo the operation.

Three structural results. Pairwise independence (Section 6): no one of the three capabilities implies any other. Lean-verified finite counterexamples at sizes $N \in \{4, 5, 6, 8, 10\}$ establish all six non-implications; four are tight. Joint irredundance (Section 6.1): all eight Boolean combinations of (S, D, C) are realised by Lean-verified witnesses, so no capability is a Boolean function of the other two. Optimal coexistence (Section 7): the smallest cardinality at which all three capabilities can hold simultaneously is $N = 5$, and this is tight: ICP requires three pairwise-distinct core elements, forcing $|S| \geq 5$. Each Cayley table is found by SAT search and independently verified in Lean 4 [9] with zero sorry.

What follows from binary judgment. Several classical algebraic identities are incompatible with this setting. The proofs share a single schema: any equation whose specialisation at a classifier τ forces τ 's row to be constant collides with extensionality, equating τ with one of the truth values. From D alone: commutativity, mediality, right self-distributivity all fail (Theorems 3.4, 3.5, 3.6). From S+D: associativity, left self-distributivity, right identity all fail (Theorems 3.10, 3.11, 3.9). The classical algebraic hierarchy — semigroups, monoids, groups, rings — therefore cannot host all three capabilities, despite each capability being individually expressible in classical terms. The intuition behind the schema is that binary judgment is intrinsically non-chainable: composing the operation through a classifier collapses the judgment to a fixed verdict, eliminating discriminative content.

What lies beyond. Two structural results extend the picture. The three-category decomposition $S = Z \sqcup C \sqcup N$ induced by D is an isomorphism invariant (Theorem 4.2); the unique coarsest non-trivial isomorphism-invariant partition of core is the classifier/non-classifier split (Corollary 4.3). At $N = 5$ a structure theorem (Theorem 4.8) pins the role assignments of retraction, classifier, and ICP-triple onto the three core slots with no slack: every retraction pair has $s = r$, every ICP-triple has the form (τ_1, g, τ_2) , and the only possible non-trivial automorphism is the classifier transposition $(\tau_1 \tau_2)$ (Theorem 4.11). At $N \geq 6$ strong S re-opens and the role-shape fractures (Proposition 4.12), but partial invariants survive across the transition (Propositions 4.13, 4.14).

Relation to universality programmes. Combinatory algebras (PCAs, TCAs) [10–12] and Turing categories [14] solve a different problem: full internal universality of the operation, paid for by infinite cardinality (Theorem 3.14, the K-infinity result of the PCA tradition). The setting of this paper retains finite cardinality and obtains, in exchange, a weaker form of universality — representation without a global projector, classification without a subobject classifier, composition without a total internal hom. The three capabilities S, D, C are what survives the trade. Section 8 discusses the finite-vs-infinite cardinality dichotomy in more detail; it is context for our setting, not a fact derivable inside it.

Contributions.

- (1) A positive axiomatisation of the setting as finite extensional algebras with binary judgment, together with a uniform classifier-constancy schema explaining why associativity, mediality, self-distributivity, right identity, and commutativity all fail (Section 3).
- (2) An independence theorem: S, D, and C are pairwise independent, with Lean-verified counterexamples in all six directions at $N = 4$ through $N = 10$ (Section 6), and coexistence witnesses at $N = 5$ and $N = 6$ (Section 7).
- (3) An optimal bound: $N = 5$ is the minimum carrier size for S+D+C coexistence; ICP is impossible at $N = 4$ by pigeonhole (Theorem 7.1).
- (4) Decomposition invariance and role rigidity: the $Z/C/N$ partition is preserved by isomorphisms (Section 4); the $N \geq 6$ principal coexistence witness and the smaller S+D witnesses have trivial automorphism group, while the canonical $N = 5$ S+D+C witness internalises a classifier-swap as the action

of its non-classifier (Proposition 4.6, Remark 9); at $N = 5$ a structure theorem forces retraction, classifier, and C-triple onto the three core slots and rules out strong S, and the mirror-row theorem (Theorem 4.11) bounds $|\text{Aut}(M)| \leq 2$; at $N \geq 6$ strong S returns and rigidity fails (Proposition 4.12).

- (5) A characterization of C: the ICP is logically equivalent to the standard Compose+Inert axioms under variable renaming, with Lean-verified necessity of the non-degeneracy guard conditions (Section 5).
- (6) Joint irredundance: all eight cells of the (S, D, C) Boolean cube are inhabited by Lean-verified witnesses (Section 6.1), so no capability is a Boolean function of the other two.

2 Definitions

Throughout, $(S, \cdot)$ is a finite magma on $\text{Fin}(n) = \{0, 1, \dots, n-1\}$ with binary operation $\cdot : \text{Fin}(n) \times \text{Fin}(n) \rightarrow \text{Fin}(n)$.

Definition 2.1 (Finite Extensional Algebra with Binary Judgment). A finite extensional algebra with binary judgment — equivalently, an extensional 2-pointed magma — is a tuple $(S, \cdot, z_1, z_2)$ where:

- (1) $z_1, z_2 \in S$ are distinct elements designated as the *truth values* of the algebra.
- (2) The truth values are *operationally inert*: $z_i \cdot x = z_i$ for all $x \in S$. Equivalently, z_1 and z_2 are left-absorbers.
- (3) No other element is operationally inert: no other element is a left-absorber.
- (4) *Extensionality*: distinct elements have distinct rows in the Cayley table.

We write $\text{core}(S) := S \setminus \{z_1, z_2\}$ for the elements of S that are not truth values.

The four clauses isolate four structural commitments. Clause (i) declares two distinguished judgment outputs and requires them to be distinct — “binary” means two-valued. Clause (ii) gives the truth values their operational meaning: their post-judgment behaviour depends only on their judgment status, not on what they are judging. Clause (iii) closes the truth-value subset against operational impostors: every other element of S varies under left-multiplication, so no other element “acts like a truth value” without being one. Clause (iv) is the foundational extensional commitment: an element’s row in the Cayley table is a complete operational description of it, so equality of behaviour is equality.

The choice of cardinality two for the judgment subset is forced by non-degeneracy:

Remark 1 (Why binary, not unary). If a candidate algebra had only one truth value z , any element τ purportedly classifying inputs against $\{z\}$ would satisfy $\tau \cdot x = z$ for all x . The row of τ would then equal the row of z (both constantly z), and extensionality would force $\tau = z$ — the supposed classifier collapses into the truth value itself. Binary judgment is therefore the smallest non-degenerate case, not a parameter choice; k -valued generalisations for $k \geq 3$ strengthen the theory in a different direction we do not pursue here. Lean: `single_absorber_collapse`.

The clause (iii) requirement that no other element be a left-absorber is independent of clauses (i), (ii), and (iv) together with the classifier dichotomy of Definition 2.4: at every $N \in \{4, 5, 6\}$, Z3 produces an extensional 2-pointed magma satisfying D in which a third

core element has a constant row (script scripts/canonicity/no_other_2e) is independent of (S) , either
 The clause is therefore a genuine axiom, not a consequence of the rest.

We do *not* commit to a stronger reading of the two truth values as “opposites” in any structural sense. The theorems are compatible both with a symmetric reading (two indistinguishable judgment outputs) and a polar reading (two poles of a polarity). Whether z_1 corresponds to “false” and z_2 to “true” or vice versa is a free choice the axioms do not constrain. This freedom is genuine: applications can specialise the reading without changing the theorems.

Definition 2.2 (Faithful Retract Magma). A *Faithful Retract Magma* (FRM) extends an extensional 2-pointed magma with a *retraction pair* $s, r \in \text{core}(S)$ satisfying $r \cdot (s \cdot x) = x$ for all $x \in \text{core}(S)$, with $r \cdot z_1 = z_1$ (anchoring). The remaining absorber outputs of s and r are unconstrained.

Retraction pair convention. Throughout this paper the retraction pair (s, r) satisfies the mutual-inverse condition on core: $r \cdot (s \cdot x) = x$ and $s \cdot (r \cdot x) = x$ for all $x \in \text{core}(S)$. This matches the Lean formalisation. The choice is load-bearing, not stylistic. One-sided retraction ($r \cdot (s \cdot x) = x$) is a decodable-encoding condition: s produces codes that r can recover. Mutual inverse is the stronger, faithful-representation condition: s and r invert each other in both directions on core, so the encoding neither invents nor loses information. Most theorems (9 of 11) go through under the one-sided hypothesis, and we note this where relevant; two results, retraction pair placement (Theorem 3.7) and the tight bound $|S| \geq 5$ when $s \neq r$ (Theorem 3.8), genuinely require the full mutual inverse. Proposition 2.3 shows this distinction is non-vacuous.

PROPOSITION 2.3 (ONE-SIDED / MUTUAL-INVERSE SEPARATION). *There is an extensional 2-pointed magma on $\text{Fin}(4)$ admitting a one-sided retraction pair (s, r) with $s \neq r$, $s, r \in \text{core}$, $r \cdot z_1 = z_1$, and $r \cdot (s \cdot x) = x$ for all $x \in \text{core}$, but no mutual-inverse pair (s', r') with $s' \neq r'$. The explicit Cayley table (rows indexed by the left operand) is:*

$\cdot$	0	1	2	3
0	0	0	0	0
1	1	1	1	1
2	1	0	1	3
3	0	2	1	3

with $z_1 = 0$, $z_2 = 1$, $s = 2$, $r = 3$. *Mutual inverse fails at $x = 2$: $s \cdot (r \cdot 2) = 2 \cdot 1 = 0 \neq 2$. Lean: mutual_strictly_stronger_than_one_sided.*

PROOF. All axioms are verified by direct inspection; Lean discharges each by decide, including the non-existence of a mutual-inverse pair with $s' \neq r'$ over $\text{Fin}(4) \times \text{Fin}(4)$. The separation establishes that mutual inverse strictly implies one-sided retraction, so the choice of axiom is substantive, and that the tight bound of Theorem 3.8 cannot be weakened to one-sided. $\square$

The retraction pair gives the algebra a section/retraction structure on core: s acts as a section and r as a retraction. We do not assume a canonical global encoding.

Definition 2.4 (Classifier Dichotomy). An extensional 2-pointed magma satisfies the *classifier dichotomy* when:

- (1) A classifier $\tau \in \text{core}(S)$ exists with $\tau \cdot x \in \{z_1, z_2\}$ for all $x \in S$.

- (2) For independence of (S) , either
 - $y \cdot x \in \{z_1, z_2\}$ for all $x \in \text{core}(S)$ — y is a *classifier*, or
 - $y \cdot x \notin \{z_1, z_2\}$ for all $x \in \text{core}(S)$ — y is a *non-classifier*.
- (3) At least one non-classifier exists (the dichotomy is not one-sided).

The three clauses split into structural and non-vacuity content. Clauses (i) and (ii) together, D_{struct} , are the structural fragment: the algebra contains *some* classifier and the classification is global. Clause (iii) ensures both classes are inhabited. In the joint S+D+C setting, clause (iii) is automatic: the middle element b of any ICP triple is core-preserving (Definition 2.5 clause 1), hence a non-classifier, hence supplies the missing witness (Lean: `DStructRetractMagma.has_non_classifier_of_icp`). Theorem 3.3 below makes the split formal: $D \iff D_{\text{struct}} \wedge P$, where P asserts the existence of a core-preserving element.

The dichotomy creates a three-category decomposition $S = Z \sqcup C \sqcup N$ where $Z = \{z_1, z_2\}$ (truth values), C (classifiers), and N (non-classifiers). No element can be partially classifying. We call outputs in $\{z_1, z_2\}$ *absorber-valued*.

Definition 2.5 (Internal Composition Property). A magma $(S, \cdot, z_1, z_2)$ satisfies the *Internal Composition Property* (ICP) when:

$\exists a, b, c \in \text{core}(S)$, pairwise distinct, such that:

- (1) *Core-preservation*: $b \cdot x \notin \{z_1, z_2\}$ for all $x \in \text{core}(S)$.
- (2) *Factorization*: $a \cdot x = c \cdot (b \cdot x)$ for all $x \in \text{core}(S)$.
- (3) *Non-triviality*: $|\{a \cdot x : x \in \text{core}(S)\}| \geq 2$.

ICP says that the left-regular representation contains a non-trivial factorization on core: one core action factors through two others, one of them core-preserving — a single fragment of internal composition. The pairwise-distinctness and non-triviality clauses are guard conditions blocking degeneracies; both are proved necessary by Lean-verified witnesses (Section 5). In Section 5 we prove that ICP is equivalent to the standard operational axioms *Compose+Inert* (Theorem 5.1).

Definition 2.6 (The Three Capabilities). An extensional 2-pointed magma has:

- Capability S (*internal splitting*) if it admits a retraction pair (Definition 2.2).
- Capability D (*internal dichotomy*) if it satisfies the classifier dichotomy (Definition 2.4).
- Capability C (*internal composition*) if it satisfies the Internal Composition Property (Definition 2.5).

All three capabilities are properties of the same base structure. None is defined as an extension of any other, making the independence theorem a statement about six non-implications, all proved by Lean-verified counterexamples.

Why these three. Each capability answers a specific positive question about a finite extensional algebra with binary judgment, in a way that is operational rather than typed. Extensionality identifies each element with its row in the Cayley table, so the elements of the algebra are a family of typed maps $\{L_a^{\text{core}} : \text{core} \rightarrow \text{core} \cup Z\}$, one per element, with the truth-value maps L_{z_1}, L_{z_2} being constant. S , D , and C correspond to three fundamental structural questions about such a family:

- *Splitting* (S): does the identity on core split through the family — do some L_s, L_r satisfy $L_r \circ L_s = \text{id}_{\text{core}}$? This is the standard section-retraction pair [3].
- *Codomain purity* (D): does each map hit core or Z , never both? This is the unique coarsest non-trivial isomorphism-invariant partition of core (Corollary 4.3).
- *Factorization* (C): does the family contain a non-trivial composition $L_a = L_c \circ L_b$ with L_b core-preserving and L_a non-constant? This is the weakest factorization condition that is not vacuously satisfiable (Lean-verified separations in ICP.lean).

Each property is minimal in its own sense. S is the standard notion of splitting. D is the coarsest invariant partition. C is the weakest non-degenerate factorization. The independence and joint-irredundance results (Theorems 6.1, 6.2) confirm these are three questions, jointly irredundant.

Other structural questions. The slice (S, D, C) is one choice within a broader axiom landscape. A Z3 canonicity probe (scripts/canonicity/probe_axioms.py) tested thirteen candidate operational axioms on extensional 2-pointed magmas at $N \in \{5, 6\}$ via pairwise directional implication queries against S, D, C; ten are pairwise independent of (S, D, C) in both directions at both sizes, three collapse into derived consequences (retraction pair gives one-sided retraction and left-cancellation; both D and C imply P via Theorem 3.3). (S, D, C) is therefore one structurally selectable slice among many. What distinguishes it is that the canonical $N = 5$ S+D+C witness is fully determined by the topos-theoretic indicator reading of D together with the requirement that the magma's automorphism be internalised in its operation (Remark 9); the other ten probe axioms have no such joint structural characterisation.

The internalisation frame. The magma operation $\cdot$ is taken as given throughout — the structuring relation of $(A, \cdot)$, not itself an object inside the structure. This is not a gap but a structural commitment: no finite algebra can contain its own operation as an element without generating a meta-operation, a Tarski/Lawvere-style regress. We study what *can* be internalised within $(A, \cdot)$ modulo $\cdot$: services that would otherwise be supplied externally (naming and decoding, binary judgment, operation chaining), realised instead as behaviours of elements relative to the fixed operation. The three capabilities S, D, C are each one such internalisation; the operation $\cdot$ is the one component deliberately left outside the internalisation catalog.

3 Universal Theorems

The following theorems are proved by pure algebraic reasoning (no decide) and formalized in `Dichotomic.lean` and `NoCommutativity.lean`. We state the hypotheses explicitly: some require only D, while others use both S and D.

3.1 Theorems from D alone

THEOREM 3.1 (THREE-CATEGORY DECOMPOSITION). *In any extensional 2-pointed magma satisfying D, every element is a zero, a classifier, or a non-classifier. The classes are pairwise disjoint: in particular, $C \cap N = \emptyset$ (no element is partially classifying).*

THEOREM 3.2 (P-WITNESS IDENTITY UNDER D). *In any extensional 2-pointed magma satisfying D, an element $y \in \text{core}(S)$ is a P-witness if and only if y is a non-classifier.*

PROOF. ($\Rightarrow$) Let y be a P-witness, so $y \cdot x \in \text{core}(S)$ for all $x \in \text{core}(S)$. Since $\text{core}(S) \cap \{z_1, z_2\} = \emptyset$, we have $y \cdot x \notin \{z_1, z_2\}$ for all $x \in \text{core}(S)$. Hence y is a non-classifier.

($\Leftarrow$) Let y be a non-classifier, so $y \cdot x \notin \{z_1, z_2\}$ for all $x \in \text{core}(S)$ by Definition 2.4. Since $y \cdot x \in S = \{z_1, z_2\} \cup \text{core}(S)$ and $y \cdot x \notin \{z_1, z_2\}$, we have $y \cdot x \in \text{core}(S)$. Hence y is a P-witness. $\square$

Remark 2. Theorem 3.2 equates the P-witness set with the non-classifier set, so the count of P-witnesses equals the count of non-classifiers in any magma satisfying D. At $N = 5$ under S+D+C, the structure theorem (Theorem 4.8) forces exactly one non-classifier, so the P-witness set has size 1, and that single element absorbs all P-related roles — it is simultaneously s , r , and the middle element b of the C-triple. At $N \geq 6$ the non-classifier count can be ≥ 2 (Proposition 4.12), so P-witnesses distribute across multiple elements and role separation becomes available. This identifies one structural mechanism behind the $N = 5 \rightarrow N = 6$ phase transition.

THEOREM 3.3 (D-FACTORIZATION THROUGH P). *Let D_{struct} denote the conjunction of dichotomy on $\text{core}(S)$ and existence of a full classifier τ (i.e., D minus the non-classifier-existence clause). For any extensional 2-pointed magma at any finite N, $D \iff D_{\text{struct}} \wedge P$.*

PROOF. Both directions are direct corollaries of Theorem 3.2.

($\Rightarrow$) D gives D_{struct} together with a non-classifier y . By Theorem 3.2, y is a P-witness, so P holds.

($\Leftarrow$) D_{struct} supplies dichotomy on $\text{core}(S)$ and a full classifier τ ; P supplies a witness $y \in \text{core}(S)$ with $y \cdot x \in \text{core}(S)$ for all $x \in \text{core}(S)$. By Theorem 3.2, y is a non-classifier. Hence all three D clauses hold. $\square$

THEOREM 3.4 (ASYMMETRY). *No extensional magma with two distinct left-absorbers is commutative.*

THEOREM 3.5 (NO MEDIALITY). *No extensional 2-pointed magma carrying a classifier is medial: the identity $(a \cdot b) \cdot (c \cdot d) = (a \cdot c) \cdot (b \cdot d)$ cannot hold universally. Setting $a = c = \tau$ and $b = d$ collapses both sides via the classifier-plus-absorber identity, yielding $\tau \cdot b = \tau \cdot \tau$ for all b ; extensionality then forces $\tau \in \{z_1, z_2\}$. Lean: no_mediality.*

THEOREM 3.6 (NO RIGHT SELF-DISTRIBUTIVITY). *No extensional 2-pointed magma carrying a classifier satisfies right self-distributivity: the identity $(a \cdot b) \cdot c = (a \cdot c) \cdot (b \cdot c)$ cannot hold universally. The proof instantiates the same pattern as Theorem 3.5: with $a = \tau$, both sides reduce to absorber values under τ 's classifier property, forcing τ 's row constant. Lean: no_right_self_distributivity.*

3.2 Theorems from S+D

THEOREM 3.7 (RETRACTION PAIR PLACEMENT). *In any extensional 2-pointed magma satisfying both S and D, any witness r to S is a non-classifier: $r \in N$. (Under the stronger mutual-inverse condition in the Lean formalization, $s \in N$ also holds.)*

THEOREM 3.8 (CARDINALITY BOUNDS). *$|S| \geq 4$ for any extensional 2-pointed magma satisfying both S and D. (The bound $|S| \geq 4$ is nearly immediate from the definitions, since the four roles (two absorbers,*

one classifier, one non-classifier) must be distinct; the Lean-verified proof confirms that no subtle interaction reduces it. The bound is tight: a 4-element witness with $s = r$ is exhibited in Appendix A.) Under the stronger mutual-inverse condition with $s \neq r$, $|S| \geq 5$ (also tight); this is the nontrivial bound, as it shows the retraction pair consumes two distinct core slots.

THEOREM 3.9 (NO RIGHT IDENTITY). *No extensional 2-pointed magma satisfying both S and D has a right identity element.*

THEOREM 3.10 (NO ASSOCIATIVITY). *No extensional 2-pointed magma carrying both a classifier and a retraction pair is associative. In particular, no extensional 2-pointed magma satisfying both S and D is a semigroup, monoid, group, or ring; the classifier dichotomy with a retraction pair is incompatible with the entire classical algebraic hierarchy.*

PROOF. Assume $(\tau \cdot a) \cdot b = \tau \cdot (a \cdot b)$ for all a, b . We derive a contradiction in four steps. The proof uses only extensionality, the existence of a classifier τ , and a retraction pair (s, r) . Neither the global dichotomy nor the non-classifier existence axiom is needed. Lean file: Dichotomic.lean.

Step 1: τ absorbs on the right. For any a, b :

$$\tau \cdot (a \cdot b) = (\tau \cdot a) \cdot b = z_i \cdot b = z_i = \tau \cdot a$$

where the second equality uses $\tau \cdot a \in \{z_1, z_2\}$ (classifier), and the third uses left-absorption. So $\tau \cdot (a \cdot b) = \tau \cdot a$ for all a, b .

Step 2: τ is constant on core. For any core x : the retraction pair gives $x = r \cdot (s \cdot x)$. Applying Step 1 with $a = r$, $b = s \cdot x$:

$$\tau \cdot x = \tau \cdot (r \cdot (s \cdot x)) = \tau \cdot r.$$

Set $v = \tau \cdot r \in \{z_1, z_2\}$. Then $\tau \cdot x = v$ for all core x .

Step 3: τ is constant on absorbers. Since τ is a classifier, it hits both z_1 and z_2 ; otherwise, if $\tau \cdot x = z_1$ for all x , then τ and z_1 have identical rows, so $\tau = z_1$ by extensionality, contradicting $\tau \neq z_1$ (and symmetrically for z_2). Let b_1, b_2 witness $\tau \cdot b_1 = z_1$ and $\tau \cdot b_2 = z_2$. Applying Step 1 with $a = \tau$:

$$\tau \cdot z_1 = \tau \cdot (\tau \cdot b_1) = \tau \cdot \tau = v, \quad \tau \cdot z_2 = \tau \cdot (\tau \cdot b_2) = \tau \cdot \tau = v.$$

The middle equalities use Step 1; $\tau \cdot \tau = v$ because τ is core (Step 2).

Step 4: Contradiction. Steps 2 and 3 give $\tau \cdot x = v$ for all $x \in S$. Since $v \in \{z_1, z_2\}$, the classifier τ has the same row as absorber v . By extensionality, $\tau = v \in \{z_1, z_2\}$, contradicting $\tau \neq z_1, z_2$. $\square$

Remark 3. The retraction pair is essential in Step 2: the identity $\tau \cdot x = \tau \cdot (r \cdot (s \cdot x)) = \tau \cdot r$ uses $r \cdot (s \cdot x) = x$ to show τ is constant on core. Without a retraction pair, the proof breaks, and the theorem does not extend to “no extensional 2-pointed magma with a classifier is associative.”

THEOREM 3.11 (NO LEFT SELF-DISTRIBUTIVITY). *No extensional 2-pointed magma satisfying both S and D satisfies left self-distributivity: the identity $a \cdot (b \cdot c) = (a \cdot b) \cdot (a \cdot c)$ cannot hold universally. The specialization $a = \tau$ yields $\tau \cdot (b \cdot c) = \tau \cdot b$ (Step 1 of the no-associativity proof); the retraction-pair argument (Steps 2–4) then proceeds identically. Lean: no_left_self_distributivity.*

The classifier-constancy schema. Theorems 3.4, 3.5, 3.6, 3.10, and 3.11 share a proof schema, summarised in Section 1: specialisation at $a = \tau$ (or $a = c = \tau$) forces τ ’s row constant, colliding with extensionality. Commutativity, mediality, and right self-distributivity instantiate the schema from D alone; associativity and left self-distributivity use $S+D$, with the retraction pair extending classifier-constancy from core to absorbers.

3.3 Theorems from D+C

THEOREM 3.12 (H-TRIPLE TYPE-COHERENCE). *In any extensional 2-pointed magma satisfying both D and C , the outer elements a and c of any H-triple (a, b, c) are either both classifiers or both non-classifiers.*

PROOF. Let (a, b, c) be an H-triple, so $a \cdot x = c \cdot (b \cdot x)$ for all $x \in \text{core}(S)$, with $b \cdot x \in \text{core}(S)$ for all $x \in \text{core}(S)$ (core-preservation of b).

Suppose c is a classifier. Then $c \cdot y \in \{z_1, z_2\}$ for all $y \in \text{core}(S)$, so in particular $c \cdot (b \cdot x) \in \{z_1, z_2\}$ for all $x \in \text{core}(S)$. By the H-triple equation, $a \cdot x \in \{z_1, z_2\}$ for all $x \in \text{core}(S)$, so a is also a classifier.

Suppose c is a non-classifier. Then $c \cdot y \notin \{z_1, z_2\}$ for all $y \in \text{core}(S)$, so $c \cdot (b \cdot x) \notin \{z_1, z_2\}$ for all $x \in \text{core}(S)$. By the H-triple equation, $a \cdot x \notin \{z_1, z_2\}$ for all $x \in \text{core}(S)$, so a is also a non-classifier.

Either way, a and c have the same classifier-type. $\square$

Remark 4. Theorem 3.12 is invoked at Step 3 of the $N = 5$ structure theorem (Theorem 4.8) and isolates the universal- N content that the structure theorem then specializes to size 5.

THEOREM 3.13 (CLASSIFIER-COUNT BOUND). *In any extensional 2-pointed magma satisfying both D and C , either $|C| \geq 2$ or $|N| \geq 3$, where C and N denote the classifier and non-classifier subsets of $\text{core}(S)$.*

PROOF. Let (a, b, c) be any H-triple (existing by C). By Theorem 3.2, b is a non-classifier (since b is core-preserving by clause 1 of Definition 2.5), so $b \in N$. By Theorem 3.12, a and c are either both in C or both in N .

If $a, c \in C$: since $a \neq c$, $|C| \geq 2$.

If $a, c \in N$: since a, b, c are pairwise distinct, $|N| \geq 3$. $\square$

Remark 5. At $N = 5$ the core has exactly 3 elements, so $|C| + |N| = 3$. D requires $|C| \geq 1$ (a full classifier exists), so $|N| \leq 2$, ruling out the $|N| \geq 3$ branch of Theorem 3.13. Hence $|C| \geq 2$, and combined with $|N| \geq 1$ this forces $|C| = 2$ and $|N| = 1$ exactly — recovering the classifier count of the $N = 5$ structure theorem (Theorem 4.8). At $N \geq 6$ both disjuncts become satisfiable: the $|C| \geq 2$ branch is realized by the explicit witness in Appendix A, and the $|N| \geq 3$ branch is realized by the structurally distinguished single-classifier class in the strong- S enumeration of scripts/enumerate_rdh_iso_N6_strongR.json.

3.4 Contrast with combinatory algebras: K-infinity

The following classical result, recalled here in self-contained form, frames the contrast between this paper’s finite-cardinality setting and the standard PCA/TCA tradition (cf. Section 1): combinatory

algebras pay infinite cardinality for full internal universality, while extensional 2-pointed magmas retain finiteness and obtain the partial forms S , D , C .

THEOREM 3.14 (K-INFINITY). *If $(A, \cdot, \mathbf{k}, \mathbf{s})$ is a combinatory algebra (total application satisfying $\mathbf{k} \cdot a \cdot b = a$ and the $\mathbf{s}$ equations) with $|A| \geq 2$, then A is infinite.*

PROOF. The map $\varphi(x) = \mathbf{k} \cdot x$ is injective: $\mathbf{k} \cdot a = \mathbf{k} \cdot b$ implies $\mathbf{k} \cdot a \cdot c = \mathbf{k} \cdot b \cdot c$ for all c , hence $a = b$. We claim $\mathbf{k} \notin \text{Im}(\varphi)$. Suppose $\mathbf{k} \cdot x = \mathbf{k}$ for some x . Then for all y , $\mathbf{k} \cdot x \cdot y = \mathbf{k} \cdot y$, so $x = \mathbf{k} \cdot y$ for all y . By injectivity, all y are equal, contradicting $|A| \geq 2$. Thus $\varphi : A \rightarrow A$ is an injection missing $\mathbf{k}$, which is impossible for finite A . $\square$

Remark 6. The proof uses only the $\mathbf{k}$ equation ($\mathbf{k} \cdot a \cdot b = a$) and totality; $\mathbf{s}$ is irrelevant. Our extensional 2-pointed magmas avoid this obstruction: instead of a projector $\mathbf{k}$ with $\mathbf{k} \cdot a \cdot b = a$, they have left-absorbers z with $z \cdot x = z$ (the dual pattern, constant on columns rather than rows) and a retraction pair (s, r) with $r \cdot (s \cdot x) = x$ on core. Neither structure provides a non-surjective injection on the full ground set.

Remark 7 (K-infinity and finite computational substrates). K-infinity shows that no finite carrier admits a total, global $\mathbf{k}$ combinator — the kind of universality that packs all of computation into a single element acting on the rest. It does *not* rule out the weaker notion of a finite algebra serving as the *instruction set* of a universal interpreter whose state and memory are external. A finite Cayley table can in principle function like a CPU’s instruction table: every step of a computation is a table lookup, while the evolving state (working memory, stack, continuation) lives outside the algebra as finite sequences of element-indices that grow with the input. In this bounded-memory sense of universality — analogous to a Turing machine whose transition table is finite but whose tape grows with the input — finite extensional 2-pointed magmas are not ruled out as computational substrates. The obstruction bites only when one asks for universality to be *entirely internal* to a single global element of a finite carrier; distributed, stepwise universality over a finite instruction set remains available.

Remark 8 (K-infinity and Scott’s continuous resolution). Theorem 3.14 is the finite-carrier analogue of the obstruction Scott resolved by moving to continuous lattices and domain theory [7]. Scott’s $D \cong [D \rightarrow D]$ requires a space whose elements are simultaneously values and functions on values, which is impossible at full function-space cardinality; the resolution is to restrict to continuous functions on an inverse-limit lattice, at the price of countable infinity. Scott’s construction retains global self-application and pays the price of infinite cardinality; the E2PM setting retains finite cardinality and pays the price of global self-application. The three capabilities S , D , and C are what survives: representation without a universal projector, classification without a subobject classifier, composition without a total internal hom — the weaker partial forms of the same computational services Scott’s construction provides globally.

4 Decomposition Invariance

Definition 4.1 (Absorber-Preserving Isomorphism). An absorber-preserving isomorphism $\varphi : M_1 \rightarrow M_2$ between extensional 2-pointed magmas is a bijection $\varphi : \text{Fin}(n) \rightarrow \text{Fin}(n)$ satisfying:

- $\varphi(a \cdot_1 b) = \varphi(a) \cdot_2 \varphi(b)$ for all a, b
- $\varphi(z_1^{(1)}) = z_1^{(2)}$ and $\varphi(z_2^{(1)}) = z_2^{(2)}$

THEOREM 4.2 (FUNCTORIALITY). Any absorber-preserving isomorphism between extensional 2-pointed magmas satisfying D preserves the three-category decomposition: φ maps zeros to zeros, classifiers to classifiers, and non-classifiers to non-classifiers.

PROOF SKETCH. *Zeros:* if a is a left-absorber, then for all x in the image of φ , $\varphi(a) \cdot_2 \varphi(x) = \varphi(a \cdot_1 x) = \varphi(a)$. By surjectivity, $\varphi(a)$ is a left-absorber.

Classifiers: if $y \cdot_1 x \in \{z_1, z_2\}$ for all core x , then $\varphi(y) \cdot_2 \varphi(x) = \varphi(y \cdot_1 x) \in \{z_1', z_2'\}$ for all core $\varphi(x)$. By surjectivity and injectivity (to exclude absorber preimages), $\varphi(y)$ is a classifier.

Non-classifiers: symmetric argument using injectivity to transport $\notin \{z_1, z_2\}$.

Algebraic proof (no decide); Lean file: `Functoriality.lean`. $\square$

COROLLARY 4.3 (CANONICITY OF THE DECOMPOSITION). The $Z/C/N$ partition is the unique coarsest non-trivial isomorphism-invariant partition of the core. Any absorber-preserving isomorphism preserves each class individually (Theorem 4.2), so no coarser partition of core into two non-empty parts can be invariant unless it coincides with C/N . The classification is determined by a single algebraic invariant: whether an element’s row on core has image contained in $\{z_1, z_2\}$.

THEOREM 4.4 (CAPABILITY INVARIANCE). All three properties S , D , and C are preserved by absorber-preserving isomorphisms of extensional 2-pointed magmas: if $\varphi : M_1 \rightarrow M_2$ is such an isomorphism and M_1 satisfies S (resp. D , C), then M_2 satisfies S (resp. D , C). Equivalently, S , D , and C are structural properties of the row-image partial monoid $R(M)$ — section-retraction in $R(M)$, codomain decomposition $R(M)|_{\text{core}} = R_{\text{cls}} \sqcup R_{\text{core}}$, and partial composition closure on R_{core} — and so are automatically iso-invariant. Algebraic proof in `CapabilityInvariance.lean`; the row-image translation is verified empirically across all 3901 $N=5$ $S+D+C$ iso classes (`scripts/row_image_invariants.py`).

4.1 Role Rigidity

Corollary 4.3 fixes *role classes*: Z , C , and N are intrinsic invariants of the Cayley table. Individual role assignments within each class — which absorber is z_1 , which classifier is τ , which core element is the section s versus the retraction r — remain, in general, determined only up to $\text{Aut}(M)$.

Definition 4.5 (Role-Rigid Magma). A magma $(S, \cdot)$ is *role-rigid* if the only operation-preserving bijection $\sigma : S \rightarrow S$ is the identity — equivalently, $\text{Aut}(M)$ is trivial. Role-rigidity sharpens the class-level canonicity of the $Z/C/N$ partition to the element level: individual role assignments are $\text{Aut}(M)$ -invariants of the Cayley table.

PROPOSITION 4.6 (RIGIDITY AND NON-RIGIDITY OF THE PAPER’S PRINCIPAL WITNESSES). The following witnesses constructed in this paper are role-rigid: `dotW6` (the $N=6$ $S+D+C$ witness), `dotK4` (the $N=$

4 $S+D$ witness), dotK5 (the $N=5$ $S+D$ witness with $s \neq r$), and dotOSS (the $N=4$ one-sided separator of Proposition 2.3). Each is discharged by `native_decide` over `Equiv.Perm(Fin(n))` in `Rigidity.lean`.

The $N=5$ $S+D+C$ principal witness dotW5 (Appendix A) is non-role-rigid by design: it admits the classifier transposition $(\tau_1 \tau_2)$ as a non-trivial automorphism, and the mirror-row theorem (Theorem 4.11) ensures this is its only non-trivial automorphism, so $|\text{Aut}(\text{dotW5})| = 2$. Lean: `witness5_classifier_swap_aut` in `Rigidity.lean`. The reason this witness is non-rigid is structural — explained in the canonical-witness remark below.

Remark 9 (The canonical $N=5$ witness internalises its symmetry). The $N=5$ witness dotW5 is selected by two structural principles, not engineering convenience.

Indicator-style classifiers. Reading absorbers as Boolean truth values ($z_1 = \text{FALSE}$, $z_2 = \text{TRUE}$ — which is what D is topologically, with $\Omega = 1 + 1$), each classifier τ_i is the characteristic function of the singleton subobject $\{\tau_i\}$ extended to fix the truth labels: $\tau_i(x) = z_2$ iff $x \in \{\tau_i, z_2\}$, else z_1 . Each classifier identifies itself as TRUE and preserves both absorbers' truth labels.

The non-classifier internalises the classifier swap. The unique non-classifier g acts on core as the transposition $(\tau_1 \tau_2)$, with $g \cdot g = \text{id}$ and g fixing both absorbers. The classifier-swap symmetry guaranteed by Theorem 4.11 is therefore realised inside the magma as g 's own action on core — the non-trivial automorphism isn't an external symmetry of the Cayley table, it's the map $x \mapsto g \cdot x$ on core.

These two principles together fully determine the Cayley table; this is the unique $N=5$ $S+D+C$ magma (up to absorber-preserving isomorphism) satisfying both. The cost of this choice is that the witness is non-rigid; the alternative (taking g as the identity on core) would be rigid but would leave the classifier-swap symmetry as a property of the table only, not internalised in the operation. The internalising choice was preferred because it matches the categorical-canonicity reading the paper has been pushing toward, and because it makes the mirror-row theorem's symmetry visibly realised rather than merely possible.

COROLLARY 4.7 (POLAR ABSORBERS IN ROLE-RIGID MAGMAS). *In a role-rigid magma, no automorphism exchanges the two absorbers: z_1 and z_2 lie in distinct $\text{Aut}(M)$ -orbits. Lean: `role_rigid_polar_absorbers`.*

Role-rigidity of the paper's specific witnesses is not a general consequence of $S+D+C$, even at the minimum coexistence size $N=5$. The role-*shape* (which absorber is z_i , which core element is the unique non-classifier, the canonical (τ_1, g, τ_2) form of every C -triple) is forced universally onto the $Z/C/N$ partition by Theorem 4.8; element-level rigidity is the strictly stronger condition that no non-trivial automorphism exists. The failure mode is precisely characterised by Theorem 4.11 below: at $N=5$ the automorphism group is always a subgroup of $\mathbb{Z}/2$, and a magma fails to be role-rigid exactly when the classifier transposition $(\tau_1 \tau_2)$ is itself a Cayley-table automorphism.

THEOREM 4.8 (STRUCTURE THEOREM AT $N=5$). *Let M be an extensional 2-pointed magma on $\text{Fin}(5)$ satisfying S , D , and C , with absorbers z_1, z_2 and core $C = \text{Fin}(5) \setminus \{z_1, z_2\}$ of size 3. Then:*

- (1) *The core partitions as $C = \{\tau_1, \tau_2, g\}$ where τ_1, τ_2 are classifiers and g is the unique non-classifier.*
- (2) *Every retraction pair (s, r) in M satisfies $s = r = g$.*

- (3) *Every C -triple (a, b, c) in M satisfies $b = g$ and $\{a, c\} = \{\tau_1, \tau_2\}$.*

PROOF. Write $C = \{2, 3, 4\}$. By D , every core element is either a classifier (row on C lies in $\{z_1, z_2\}$) or a non-classifier (row on C lies outside $\{z_1, z_2\}$), and at least one is a non-classifier. Write k for the number of classifiers in C ; then $k \in \{0, 1, 2\}$.

Step 1 (retraction elements are non-classifiers). Let (s, r) be any retraction pair and suppose s is a classifier. Then $s \cdot x \in \{z_1, z_2\}$ for all $x \in C$. Since $|C| = 3 > 2 = |\{z_1, z_2\}|$, pigeonhole gives distinct $x_1, x_2 \in C$ with $s \cdot x_1 = s \cdot x_2$. The retraction identity $r \cdot (s \cdot x_i) = x_i$ then forces $x_1 = x_2$, a contradiction. So s is a non-classifier. The symmetric argument using $s \cdot (r \cdot x) = x$ shows r is a non-classifier.

Step 2 (C 's middle element is a non-classifier). Let (a, b, c) be any C -triple (existing by C). By Theorem 3.2, since b is core-preserving (clause 1 of Definition 2.5), b is a non-classifier.

Step 3 (C 's outer elements have the same classifier-type). By Theorem 3.12, a and c are either both classifiers or both non-classifiers.

Step 4 (exactly 2 classifiers). By Theorem 3.13, $k \geq 2$ or the non-classifier count in core is ≥ 3 . The second alternative would fill the entire 3-element core with non-classifiers, contradicting D 's requirement that a full classifier τ exists. Hence $k \geq 2$; combined with $k \leq 2$ from the proof opening, $k = 2$.

Step 5 (role lock-in). By Step 4, $C = \{\tau_1, \tau_2, g\}$ with g the unique non-classifier. By Step 1, $s, r \in \{g\}$, so $s = r = g$. By Step 2, $b = g$. By Step 3, a and c are both classifiers; since they are distinct, $\{a, c\} = \{\tau_1, \tau_2\}$. $\square$

Theorem 4.8 is machine-checked in Lean 4 (Magma/StructureN5.lean): the uniqueness lemma `unique_non_classifier_at_N5` yields `N5_exists_unique` (clause (i)), `N5_sec_eq_ret` (clause (ii), equivalent to Corollary 4.9), and `N5_icp_triple_structure` (clause (iii)). The proof is algebraic (no `decide`, no `native_decide`) and depends only on `propext`, `Classical.choice`, `Quot.sound`. An independent check uses three Z3 UNSAT queries (negating each of (i), (ii), (iii) in turn; `scripts/structure_theo`) each query resolving in under 0.1 s.

COROLLARY 4.9 (STRONG S IS ABSENT AT $N=5$). *No $S+D+C$ magma on $\text{Fin}(5)$ admits a retraction pair with $s \neq r$. Equivalently, strong S is unsatisfiable in conjunction with $S+D+C$ at size 5.*

PROOF. By Theorem 4.8(ii), every retraction pair has $s = r$. $\square$

COROLLARY 4.10 ($N=5$ RIGIDITY UNDER STRONG S). *Every extensional 2-pointed magma on $\text{Fin}(5)$ satisfying S with $s \neq r$, D , and C is role-rigid. The statement is vacuously true by Corollary 4.9.*

THEOREM 4.11 (MIRROR-ROW THEOREM AT $N=5$). *Let M be an extensional 2-pointed magma on $\text{Fin}(5)$ satisfying S , D , and C , with classifiers τ_1, τ_2 and unique non-classifier g (Theorem 4.8). Every automorphism $\sigma \in \text{Aut}(M)$ fixes both absorbers and the unique non-classifier g , and acts on the two classifiers as either the identity or the transposition $(\tau_1 \tau_2)$. Consequently:*

- (1) $|\text{Aut}(M)| \in \{1, 2\}$;
- (2) *the unique possible non-trivial automorphism is the transposition $(\tau_1 \tau_2)$, extended by identity on $\{z_1, z_2, g\}$;*
- (3) *M is role-rigid if and only if this classifier transposition is not a Cayley-table automorphism.*

PROOF. By Theorem 4.2, σ preserves the $Z/C/N$ partition, so σ permutes $\{z_1, z_2\}$, permutes $\{\tau_1, \tau_2\}$, and fixes g (the unique non-classifier by Theorem 4.8). Hence σ lies in the Klein four-group $\text{Sym}(\{z_1, z_2\}) \times \text{Sym}(\{\tau_1, \tau_2\})$. We rule out the two cases in which σ swaps $\{z_1, z_2\}$.

Case A: σ swaps absorbers and fixes both classifiers. For any classifier τ , $\tau \cdot \tau \in \{z_1, z_2\}$. Applying σ : $\sigma(\tau \cdot \tau) = \sigma(\tau) \cdot \sigma(\tau) = \tau \cdot \tau$, so $\tau \cdot \tau$ is σ -fixed. But σ swaps $\{z_1, z_2\}$. Contradiction.

Case B: σ swaps absorbers and swaps classifiers. Since σ fixes g and g is core-preserving (clause 3 of Theorem 4.8, via Theorem 3.2), $g \cdot g \in \text{core}(S)$ is σ -fixed. Among $\{\tau_1, \tau_2, g\}$ only g is σ -fixed, so $g \cdot g = g$. The factorisation $\tau_1 \cdot x = \tau_2 \cdot (g \cdot x)$ from Theorem 4.8(iii), evaluated at $x = g$, gives $\tau_1 \cdot g = \tau_2 \cdot (g \cdot g) = \tau_2 \cdot g \in \{z_1, z_2\}$. Apply σ : $\sigma(\tau_1 \cdot g) = \sigma(\tau_1) \cdot \sigma(g) = \tau_2 \cdot g$, so $\tau_1 \cdot g$ is σ -fixed. But σ swaps $\{z_1, z_2\}$. Contradiction.

So σ fixes $\{z_1, z_2\}$ pointwise. Together with $\sigma(g) = g$, the only remaining freedom is on $\{\tau_1, \tau_2\}$, which is either fixed or swapped. $\square$

Theorem 4.11 is machine-checked in Lean 4 (N5_aut_preserves_absorbers in Magma/MirrorRow.lean, zero sorry; depends only on propext, Classical.choice, Quot.sound). The phase-cartography enumeration of 3901 $N=5$ S+D+C iso classes finds exactly 12 admitting the classifier transposition (scripts/phase_cartography_N5.json, scripts/mirror_row_conjecture.py); these are precisely the non-role-rigid cases.

PROPOSITION 4.12 (NON-RIGIDITY AT $N \geq 6$ UNDER STRONG S). *At $N = 6, 7, 8$ there exist finite extensional 2-pointed magmas satisfying S (mutual-inverse with $s \neq r$), D , and C with non-trivial automorphism group. Z3-verified witnesses with independent Python confirmation: scripts/nonrigid_rdh_N{6,7,8}_strongR.json. The $N=8$ witness admits an absorber-swap automorphism, so Corollary 4.7 does not extend to S+D+C under strong S at this size without further hypotheses.*

PROPOSITION 4.13 ($N=6$ AUTOMORPHISM CONSTRAINTS UNDER STRONG S). *Every automorphism of an extensional 2-pointed magma on $\text{Fin}(6)$ satisfying strong S , D , and C permutes $\{z_1, z_2\}$, and no such magma admits an automorphism that fixes both absorbers and has order 4 on core. Z3 UNSAT: scripts/canonicity/nonrigid_characterization.py.*

PROOF. Both claims are encoded as Z3 UNSAT queries over $\text{Fin}(6)$. The absorber-permutation constraint reduces to checking that no model exists with a non-trivial automorphism σ satisfying $\sigma(z_1) \notin \{z_1, z_2\}$; the order-4 constraint enumerates the absorber-fixing 4-cycles on core (algebraically there are six, and each is asserted not to be an automorphism of T). Each query is independent of any specific witness table. The first query is a strict proof; the second confirms that the non-rigid landscape under strong S at $N=6$ avoids one structurally distinguished cycle type. The enumeration of non-rigid iso classes at $N=6$ strong S produces a large landscape — at least 5,985 iso classes were observed before the iterative-blocking solver time-per-class grew past feasibility — dominated by absorber-fixing involutions with $|\text{Aut}| = 2$. The enumerator file scripts/enumerate_rdh_nonrigid_iso.py is the source of record; the exact total is not used elsewhere in the paper. $\square$

PROPOSITION 4.14 ($N=6$ C-TRIPLE BOUND UNDER STRONG S). *Every extensional 2-pointed magma on $\text{Fin}(6)$ satisfying strong S , D , and C admits at least two distinct C-triples $(a, b, c) \in \text{core}(S)^3$. The bound is tight: a strong-S+D+C magma on $\text{Fin}(6)$ with exactly two C-triples is Z3-constructible. Z3 verification: scripts/canonicity/structure_theorem_N6.py.*

Theorem 4.8 and Propositions 4.12–4.14 together sharpen the phase transition picture. At $N=5$ the retraction pair, the C-triple, and the classifier pair are forced onto the three core slots with no room to spare, and in particular the retraction collapses to $s = r$; strong S is not merely consistent with rigidity but outright unavailable. At $N=6$, adding one core element re-opens strong S , and with it the possibility of non-trivial automorphisms: both the classifier count and the retraction structure admit choices, and neither the structural lock-in of (i)–(iii) nor the rigidity of the principal witnesses survives. What does survive across the transition: every automorphism still permutes $\{z_1, z_2\}$, the absorber-fixing 4-cycle is forbidden, and at least two distinct C-triples persist (Propositions 4.13, 4.14). The collapse from $N=5$'s tight role lock-in to $N=6$'s flexibility is partial, not total.

A common substructure: core-preserving rows. Let P (internal preservation) denote the property: $\exists y \in \text{core}(S)$ with $y \cdot x \in \text{core}(S)$ for all $x \in \text{core}(S)$. Both D and C imply P . D supplies a non-classifier whose row on core lies entirely in core (the all-out branch of the dichotomy), so the non-classifier itself witnesses P ; C supplies the middle element b of the factorization triple, which is core-preserving by clause 1 of Definition 2.5. We name P but do not promote it to a fourth capability: because P is implied by both D and C separately, it is not pairwise-independent of them and so does not stand on the same footing as S , D , and C with each other. It is the cleanly-named common substructure they share. The two implications are qualitatively different. Writing D_{struct} for the dichotomy on core plus existence of a full classifier τ (i.e., D minus the existential “at least one non-classifier” clause), and C_{eq} for the factorization equation plus pairwise-distinctness and non-triviality (i.e., C minus core-preservation of b), Theorem 3.3 gives the D -side universally, and Z3 verification at $N \in \{5, 6\}$ gives the C -side:

$$D \iff D_{\text{struct}} \wedge P \quad (\text{any finite } N), \quad C \iff P \wedge C_{\text{eq}} \text{ but not conversely.}$$

The structural reading: P is a common substructure, but the way it sits inside D and C differs. In D it is *decoupled* from τ (the witness can be a different element); in C it is *coupled* to the factorization's middle slot b (the witness must be the same element). Verification: scripts/canonicity/factorization_D_H.py.

5 ICP Characterization

An algebra with S can encode, decode, and look up any entry in the Cayley table via the retraction pair (s, r) . But this requires an external machine to supply control flow: dispatch, composition, recursion. C internalizes a non-trivial composition pattern that would otherwise be supplied externally: dedicated elements whose left-multiplication implements composition ($\eta = \rho \circ g$) and storage (g). The ICP captures exactly this: one core action factors through two others, meaning the algebra contains a fragment of its own control flow as an element.

The standard formulation uses two operational axioms:

- *Compose*: $\eta \cdot x = \rho \cdot (g \cdot x)$ for all $x \in \text{core}(S)$

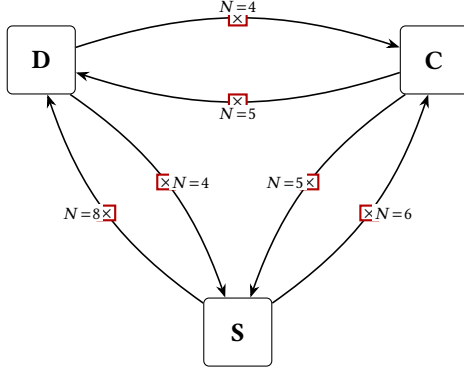


Figure 1: Full independence structure. All six pairwise non-implications are proved by Lean-verified finite counterexamples at the indicated sizes. No capability implies any other.

- *Inert*: $g \cdot x \notin \{z_1, z_2\}$ for all $x \in \text{core}(S)$

with η, g, ρ pairwise distinct non-absorbers and η non-trivial (≥ 2 distinct values on core).

THEOREM 5.1 (ICP EQUIVALENCE). *For any finite magma on a 2-pointed set $(S, \cdot, z_1, z_2)$ of any size n :*

$$\text{ICP} \iff \text{Compose} + \text{Inert (with non-trivial composite)}.$$

PROOF. The bijection between witness triples is $(\eta, g, \rho) = (a, b, c)$: the composed element a is η , the core-preserving element b is g , and the outer element c is ρ . Under this identification the factorization $a \cdot x = c \cdot (b \cdot x)$ becomes $\eta \cdot x = \rho \cdot (g \cdot x)$ (Compose), core-preservation of b becomes Inert, and the remaining conditions (pairwise distinct, non-absorber) transfer directly.

The pairwise-distinctness and non-triviality conditions are not cosmetic; they block two degeneracies, and both are proved necessary by Lean-verified witnesses (ICP.lean). Without pairwise-distinctness, the retraction pair (s, r) with $s = r$ trivially satisfies factorization via $s \cdot x = s \cdot (s \cdot x)$, giving every FRM a spurious ICP; the minimal S+D model at $N=4$ (Cayley table in Appendix A; Lean name `kripke4`) separates the weakened definition from the full one (theorem `icp_distinct_necessary`). Without non-triviality, any constant-on-core element a (with $a \cdot x = v$ for all core x) trivially factors through any core-preserving b by taking c with $c \cdot y = v$; a custom 6-element FRM separates the weakened definition from the full one (theorem `icp_nontrivial_necessary`, ICP.lean).

The proof is pure equational logic, with no decide, no case analysis on n . It holds for every finite magma on a 2-pointed set, of any size. Lean file: ICP.lean, theorem `icp_iff_composeInert`. $\square$

This equivalence gives C a single-concept definition: the left-regular representation contains a non-trivially composed element. The operational axioms (Compose, Inert) and the ICP are the same property under variable renaming. Additionally, ICP is invariant under FRM isomorphisms (algebraic proof, no decide; ICP.lean, theorem `FRMiso_preserves_icp`).

6 Independence

THEOREM 6.1 (FULL INDEPENDENCE). *No capability implies any other. Specifically, all six pairwise non-implications hold:*

- (1) $S \not\Rightarrow D$: An extensional 2-pointed magma of size 8 satisfying S contains a classifier element but fails the global dichotomy condition.
- (2) $S \not\Rightarrow C$: An extensional 2-pointed magma of size 6 satisfying S has 4 core elements (enough for ICP to be non-vacuous) but no triple satisfies the ICP.
- (3) $D \not\Rightarrow C$: An extensional 2-pointed magma of size 4 satisfying D has only 2 core elements, so ICP is vacuously unsatisfiable (pigeonhole).
- (4) $C \not\Rightarrow D$: An extensional 2-pointed magma of size 5 satisfies ICP but violates the classifier dichotomy. This is tight: ICP requires $N \geq 5$.
- (5) $D \not\Rightarrow S$: An extensional 2-pointed magma of size 4 satisfies the classifier dichotomy but admits no retraction pair. This is tight: D requires $N \geq 4$.
- (6) $C \not\Rightarrow S$: An extensional 2-pointed magma of size 5 satisfies ICP but admits no retraction pair. This is tight: ICP requires $N \geq 5$.

PROOF. Each direction is proved by exhibiting a concrete Cayley table and verifying the claimed properties by `native_decide` (or `decide` for $N \leq 8$). The tables are:

$S \not\Rightarrow D$ ($N=8$). An 8×8 Cayley table (Appendix A) defining an FRM with $s=2, r=3$. Element 4 is a classifier, but element 5's row on core is $(6, 2, 1, 1, 1, 1)$, mixing absorber and non-absorber outputs, violating the dichotomy. SAT confirms the non-implication is achievable as early as $N=3$, but vacuously: with only 1 core element, the dichotomy cannot be formulated (it requires both a classifier and a non-classifier). The $N=8$ witness is the smallest with enough core structure for D to be non-vacuously falsifiable. Lean file: `Countermodel.lean`.

$S \not\Rightarrow C$ ($N=6$, *structural*). A 6×6 Cayley table (Appendix A) defining an extensional 2-pointed magma with a retraction pair ($s=r=2$, an involution on core) but no ICP. The core has 4 elements, more than the 3 pairwise distinct witnesses ICP requires, yet no triple satisfies the ICP. As with $S \not\Rightarrow D$, the non-implication is vacuously achievable at $N=3$ (only 1 core element), but the $N=6$ witness is the smallest where ICP failure is structural rather than dimensional. Lean file: `E2PM.lean`, theorem `s_not_implies_icp_structural`.

$D \not\Rightarrow C$ ($N=4$, *tight*). The minimal S+D model at $N=4$ (`kripke4` in Appendix A) has only 2 core elements. ICP requires 3 pairwise distinct non-absorbers, so it is vacuously unsatisfiable. Lean file: ICP.lean, theorem `kripke4_no_icp`. The $N=10$ D+S model in `Countermodels10.lean` complements this: it has 8 core elements (more than enough for ICP to be non-vacuous), yet no triple satisfies ICP. The formal non-implication holds at $N=4$; the $N=10$ witness shows it persists even when C has room to exist.

$C \not\Rightarrow D$ ($N=5$, *tight*). A 5×5 Cayley table satisfying ICP but with no classifier: all three core elements map core to core (no element produces only absorber-valued outputs on core). The bound is tight: ICP requires $N \geq 5$. Lean file: `E2PM.lean`, theorem `h_not_implies_d_tight`.

$D \Rightarrow S$ ($N = 4$, *tight*). A 4×4 Cayley table with the classifier dichotomy (classifier at index 2, non-classifier at index 3) but no retraction pair among the 4 possible assignments from core. The bound is tight: D requires $N \geq 4$. Lean file: E2PM.lean, theorem d_not_implies_s_tight.

$C \Rightarrow S$ ($N = 5$, *tight*). A 5×5 Cayley table satisfying ICP but admitting no retraction pair. The bound is tight: ICP requires $N \geq 5$. Lean file: E2PM.lean, theorem h_not_implies_s_tight.

All tables are generated by Z3 SAT solver and independently verified. The Cayley tables are frozen in the repository for reproducibility. $\square$

Methodology and minimality. Each counterexample was found by encoding the relevant capability axioms (and the negation of the target capability) as a SAT problem over an $n \times n$ Cayley table, then solving with Z3, incrementing n from 3 upward. Once found, each table was frozen and independently verified in Lean. Four of six directions have provably tight bounds: $D \Rightarrow S$ and $D \Rightarrow C$ are tight at $N = 4$ (the minimum where D is non-vacuous, resp. where ICP is dimensionally impossible); $C \Rightarrow S$ and $C \Rightarrow D$ are tight at $N = 5$ (the minimum where ICP can be stated). The remaining two ($S \Rightarrow D$ at $N = 8$, $S \Rightarrow C$ at $N = 6$) are the smallest non-vacuous witnesses found. The SAT search was exhaustive: all smaller sizes returned UNSAT for every role assignment tested. Whether different role assignments or overlapping encodings could yield tighter witnesses is open.

Scaling. The independence is not a small-size artifact. For the co-existence direction, $S+D+C$ is realized at every $N \geq 5$ by an explicit construction (Theorem 7.3), so existence at all sufficiently large sizes is no longer empirical. For the six non-implication directions, we rely on SAT search: it confirms that all six are satisfiable at every size from their respective minima through $N = 15$ (the solver limit), and no size was found where any direction becomes UNSAT. Whether each non-implication holds at all $n \geq$ its minimum (i.e. a witness exists at size n for every n) remains a conjecture supported by this data, pending a constructive scaling argument analogous to Theorem 7.3.

Interpretation. The independence result shows that S, D, and C are genuinely three things, not one. An algebra can encode and decode without classifying (S without D). An algebra can classify without encoding (D without S). An algebra can internally compose without classifying or encoding (C without S or D). No capability algebraically forces any other. In particular, the $C \Rightarrow D$ direction shows that an algebra can internalize a non-trivial composition pattern without satisfying the classifier dichotomy: internal composition does not force organizational separation.

6.1 Joint irredundance: the Boolean cube

Pairwise independence shows no capability implies any other. A stronger form of irredundance asks whether some capability is a *Boolean function* of the other two: if S were equivalent to $D \wedge \neg C$, say, the triple (S, D, C) would collapse to the pair (D, C) . We rule this out by exhibiting a witness in each of the eight cells of the (S, D, C) Boolean cube.

THEOREM 6.2 (JOINT IRREDUNDANCE). *Each of the eight Boolean combinations of (S, D, C) is realized by a Lean-verified finite extensional 2-pointed magma.*

Cell	Profile	N	Witness
1	$S \wedge D \wedge C$	5	witness5
2	$S \wedge D \wedge \neg C$	4	kripke4
3	$S \wedge \neg D \wedge C$	10	hNotD
4	$S \wedge \neg D \wedge \neg C$	3	frm3 (new)
5	$\neg S \wedge D \wedge C$	5	cell15 (new)
6	$\neg S \wedge D \wedge \neg C$	4	dNoS4
7	$\neg S \wedge \neg D \wedge C$	5	hNoD5
8	$\neg S \wedge \neg D \wedge \neg C$	3	cell18 (new)

Five cells are populated by witnesses that already appear in the pairwise-independence proof, after adding the missing capability annotations. Three required fresh constructions: cell 4 and cell 8 are minimal ($N=3$) FRM and E2PM witnessing “S alone” and “no capability” respectively; cell 5 (D and C without S) is the cube’s most structurally constrained corner, an $N = 5$ magma with two classifiers, one core-preserving non-classifier serving as the ICP’s b , and no retraction pair. All verifications are machine-checked by `decide` or `native_decide` in `BooleanCube.lean`.

The cell sizes reveal an asymmetry: every cell containing C requires $N \geq 5$, while cells with only S and/or D are realized at $N = 3$ or 4. C carries strictly more structural weight than S or D, consistent with the row-scarcity argument in Section 9: ICP’s factorization demand $a \cdot x = c \cdot (b \cdot x)$ constrains three rows simultaneously and requires three pairwise-distinct core elements, which S and D do not.

Consequence. If any capability were Boolean-equivalent to a combination of the other two, some cell would be empty. All cells are inhabited: no reduction of three predicates to two exists. S, D, and C therefore carry mutually irreducible information — jointly irredundant, not merely pairwise independent.

7 Coexistence

THEOREM 7.1 (NO ICP AT $N=4$). *No extensional 2-pointed magma of size 4 satisfies the ICP. The core $\{2, 3\}$ has only 2 elements, but ICP requires 3 pairwise distinct non-absorbers. Lean: Witness5.lean, no_icp_at_4.*

THEOREM 7.2 (COEXISTENCE WITNESS). *The three capabilities S, D, and C are simultaneously satisfiable. A Lean-verified witness exists at $N=5$, which is optimal by Theorem 7.1.*

PROOF. A concrete 5×5 Cayley table is exhibited with $s = r = 2$ (identity on core), classifiers $\tau \in \{3, 4\}$, and ICP witnessed by $a = 3, b = 2, c = 4$. The retraction pair collapses to a single element ($s = r$) which doubles as the ICP’s core-preserving map; since b is the identity, the factorization $a \cdot x = c \cdot (b \cdot x)$ reduces to $a \cdot x = c \cdot x$ on core, yielding two classifiers that agree on core but distinguished by their absorber columns. The lower bound is pigeonhole: at $N = 4$, the core $\{2, 3\}$ has only 2 elements, but ICP requires 3 pairwise distinct non-absorbers (`no_icp_at_4`). Lean file: Witness5.lean, theorem sdh_witness_5. A 6×6 witness with $s \neq r$ is also verified (Witness6.lean). $\square$

THEOREM 7.3 (SCALING: $S+D+C$ EXISTS AT ALL $N \geq 5$). *For every integer $N \geq 5$, there exists an extensional 2-pointed magma on*

$\text{Fin}(N)$ satisfying S , D , and C . Lean: `WitnessAllN.lean`, *theorem* `sdh_witness_all_N`.

PROOF (CONSTRUCTIVE). For $N \geq 5$, define $T_N : \text{Fin}(N) \times \text{Fin}(N) \rightarrow \text{Fin}(N)$ by:

- $T_N(0, x) = 0$ for all x .
- $T_N(1, x) = 1$ for all x .
- $T_N(2, x) = x$ for all x .
- $T_N(3, x) = 1$ if $x \in \{3, 4\}$, else $T_N(3, x) = 0$.
- $T_N(4, x) = 1$ if $x \in \{1, 3, 4\}$, else $T_N(4, x) = 0$.
- For $k \in \{5, \dots, N-1\}$: $T_N(k, 2) = k$, $T_N(k, k) = 2$, and $T_N(k, x) = x$ for all $x \notin \{2, k\}$.

E2PM axioms. Rows 0 and 1 are constant absorbers. For $k \in \{2, 3, 4\}$ row k takes at least two values, so no row beyond rows 0, 1 is constant. Extensionality: rows 0, 1 differ from each other and from all higher rows at column 0 (row 0 has 0, row 1 has 1, row $k \geq 2$ has 0). Row 2 differs from rows 3, 4 at column 2 (row 2 has 2, rows 3, 4 have 0). Rows 3 and 4 differ at column 1. For $k, k' \in \{5, \dots, N-1\}$ with $k \neq k'$: rows k and k' differ at column k (row k has 2, row k' has k).

S . Take $s = r = 2$. Anchoring $T_N(2, 0) = 0$ holds. The retraction equations $T_N(2, T_N(2, x)) = T_N(2, x) = x$ on $\text{core}(S)$ hold trivially since row 2 is identity.

D . Rows 3 and 4 have values in $\{0, 1\}$ everywhere, so they are full classifiers. Rows 2 and k for $k \geq 5$ map $\text{core}(S)$ into $\text{core}(S)$ (verified by inspection), so they are non-classifiers. The dichotomy holds. Take $\tau = 3$ as the full classifier; the non-classifier 2 exists.

C . The triple $(3, 2, 4)$ is an H-triple: the three indices are pairwise distinct in $\text{core}(S)$; row 2 is core-preserving; for $x \in \text{core}(S)$, $T_N(3, x) = T_N(4, T_N(2, x))$ holds because row 2 is identity, so $T_N(3, x) = T_N(4, T_N(2, x)) = T_N(4, x)$ reduces to row equality on core, which holds by direct comparison (both rows are constant 0 on $\text{core} \setminus \{3, 4\}$ and constant 1 on $\{3, 4\}$); $T_N(3, \cdot)|_{\text{core}(S)}$ takes both values 0 and 1 (at $x = 2$ and $x = 3$ respectively), so non-triviality holds. $\square$

PROPOSITION 7.4 (SEPARATED-ROLE WITNESS). *There is a witness at $N=10$ in which the principal $S+D+C$ roles (2 absorbers, 2 retraction pair, 1 classifier, 3 ICP witnesses) occupy 8 distinct elements.*

PROOF. A concrete 10×10 Cayley table (Appendix A) satisfies all three capabilities with distinct role assignments. Lean file: `Witness10.lean`. A witness with $s \neq r$ (non-degenerate retraction) is exhibited at $N=6$ (Appendix A). $\square$

Minimum cardinality. $N=5$ is the optimal minimum for $S+D+C$ coexistence (Lean-verified; `Witness5.lean`). The witness uses $s = r$ (identity on core), two classifiers, and one non-classifier: the retraction pair member doubles as the ICP's core-preserving element. At $N=4$ the core has only 2 elements but ICP requires 3 pairwise distinct non-absorbers, so $N=4$ is impossible (`no_icp_at_4`).

The $N=5$ witness reveals how thin the three properties can become simultaneously. The retraction pair is the identity on core ($s = r$, no actual encoding work), and the ICP factorization $a \cdot x = c \cdot (b \cdot x)$ collapses to $a \cdot x = c \cdot x$ (row equality on core between two classifiers differing only on absorber columns). The three capabilities are formally present but barely interacting. The $N=6$ witness with $s \neq r$ (`Witness6.lean`) is the more informative example: the retraction pair does genuine encoding, the ICP factorization

is non-degenerate, and the roles do not trivially overlap. We retain the general definitions (allowing $s = r$) since the algebraic results (independence, injectivity, no-associativity) hold regardless.

8 Related Work

Partial and total combinatory algebras. PCAs [10, 11] and TCAs [12] provide the standard algebraic framework for realizability and computability. PCA theory has a rich structural theory: morphisms between PCAs, the lattice of PCA quotients, and totalizability [13] all involve detailed analysis of the application operation. PCA theory does study which elements realize which functions. However, this analysis concerns representability and inter-PCA relationships, not the *partition* of elements into behavioral types by their row structure. This question requires a finite total operation table to formulate; it does not arise in the PCA/TCA setting, where application is partial or the carrier is infinite. Bethke and Klop [13] proved that some PCAs cannot be extended to total ones, a structural result about the relationship between partiality and totality. The K-infinity theorem (our Theorem 3.14) shows that nontrivial $S+K$ algebras must be infinite, explaining why finite algebras fall outside this framework and why our setting requires a different combinator basis.

Turing categories. Cockett and Hofstra [14] define a Turing category as a cartesian restriction category with a Turing object A and a Turing morphism $\bullet : A \times A \rightarrow A$ that universally simulates all morphisms in the category. The Structure Theorem establishes that PCAs and Turing categories are two views of the same concept. Turing objects have cardinality “one or infinite” [14], ruling out nontrivial finite Turing objects. The Turing morphism is described as “the heart of the whole structure” but its algebraic properties (row structure, element classification, associativity constraints) are not studied. Where Turing categories distribute computational roles across distinct objects (programs, data, evaluators), our finite algebras develop the same roles internally as emergent behavioral regions within a single carrier set: absorbers encode failure, classifiers provide absorber-valued judgment, and the retraction pair provides encoding.

Self-interpretation. Mogensen [1] constructs self-interpreters for the λ -calculus. Brown and Palsberg [2] show that self-interpretation is possible in strongly-normalizing typed languages, contradicting earlier impossibility results. These works study self-interpretation (our S) but do not address the independence of S from D or C .

Independence in finite algebra. Independence and non-implication results between equational properties have a long history in universal algebra, from Birkhoff's characterization of equational classes [6] to Mal'cev conditions characterizing implications between properties of varieties. Our results operate at a different level: we separate *operational* properties (retraction pairs, classifier partitions, factorizations in the left-regular representation) rather than equational identities, and our base class, the finite extensional 2-pointed magmas, is not a variety in the Birkhoff sense (extensionality is a quasi-identity, not an equation). To our knowledge, the pairwise independence of retraction pairs, classifier dichotomies, and factorization properties in finite magmas has not been previously established. The SAT-find-then-Lean-verify methodology is related

to the use of automated tools such as Mace4 [8] for counterexample search in algebra, though we use Z3 for constraint solving and Lean 4 for independent formal verification.

Finite model theory and machine-checked proofs. Independence results via finite counterexamples are standard in finite model theory [5]. Each Cayley table is frozen in the Lean source and verified by the kernel, eliminating transcription errors. The sizes involved ($N=5$ through $N=10$) are small enough for `decide/native_decide` but large enough that hand-verification of all conditions would be error-prone. The universal algebraic theorems (no-associativity, decomposition invariance, ICP equivalence) are proved by pure equational reasoning without `decide`, and hold for all finite sizes.

Cellular automata and applicative calculus. The structures studied here sit at an intersection of two computational-substrate traditions that have otherwise remained separate. Cellular automata (Rule 110, Conway’s Game of Life, and the broader tradition) establish universality from a finite local rule by iterated unfolding; the rule itself is small and fixed, while computation emerges from temporal-spatial extent. Applicative calculi (combinatory logic, the λ -calculus) take the binary application operation as primitive and use extensionality to individuate functions by behavior. Finite extensional 2-pointed magmas inherit the finite-local-rule nature of cellular automata (the Cayley table is the rule; computation unfolds as a term tree) together with the applicative-and-extensional structure of combinatory calculi (the operation is $a \cdot b$, asymmetric between operator and operand, with rows individuating elements). They differ from each tradition in specific ways: from cellular automata, they drop the spatial grid, replacing it with term-tree unfolding; from combinatory calculi, they drop abstraction, fixing the entire function inventory as the finite carrier. Neither tradition has systematically studied the resulting hybrid – finite applicative substrates with extensional individuation – and in particular the capability-decomposition questions (what can be internalized as element behavior modulo the operation?) answered by this paper do not arise naturally in either parent.

9 Discussion

The classifier dichotomy is organizational. D organizes elements into coherent categories: classifiers that test and non-classifiers that transform. The $C \not\Rightarrow D$ result shows that an algebra can internalize composition without this organization. D contributes semantic structure rather than compositional behavior.

The role of extensionality. Extensionality (distinct elements have distinct rows) is load-bearing throughout. Without it, the no-associativity proof fails: Step 4 derives $\tau = v$ from identical rows. Without it, any constant-row element trivially satisfies both classifier and non-classifier conditions, making the three-category decomposition degenerate. Extensionality is not merely a convenience but the condition that makes element-level classification meaningful. An element’s identity is its row in the Cayley table; extensionality formalizes this.

Why these three properties? Extensionality identifies each element with its row in the Cayley table. The elements of a finite extensional 2-pointed magma are therefore a family of typed maps

$\{L_a^{\text{core}} : \text{core} \rightarrow \text{core} \cup Z\}$, one per element, with the absorber maps L_{z_1}, L_{z_2} being constant. S, D, and C correspond to the three fundamental structural questions about such a family:

- *Splitting (S)*: does the identity on core split through the family? That is, do some L_s, L_r satisfy $L_r \circ L_s = \text{id}_{\text{core}}$? This is a section-retraction pair [3].
- *Codomain purity (D)*: does each map hit core or Z , never both? This is the unique coarsest non-trivial isomorphism-invariant partition of core (Corollary 4.3).
- *Factorization (C)*: does the family contain a non-trivial composition, $L_a = L_c \circ L_b$ with L_b core-preserving and L_a non-constant? This is the weakest factorization condition that is not vacuously satisfiable: weakening either the pairwise-distinctness or the non-triviality requirement makes every retraction-equipped magma satisfy it trivially (Lean-verified separations in ICP.lean).

Each property is minimal in its own sense. S is the standard notion of splitting. D is the coarsest invariant partition. C is the weakest non-degenerate factorization. The independence and joint-irredundance results (Theorems 6.1, 6.2) confirm these are three questions jointly irredundant, not reducible to two. The precise categorical status of D and C is discussed further in the categorical connections paragraph below; whether either corresponds to a known universal property remains open.

(S, D, C) as a structurally distinguished slice. A Z3 canonicality probe (`scripts/canonicity/probe_axioms.py`) tested thirteen candidate operational axioms on E2PM at $N \in \{5, 6\}$ via pairwise directional implication queries against S, D, C; ten are pairwise independent of (S, D, C) in both directions at both sizes, three collapse (S subsumes one-sided retraction and left-cancellation; both D and C imply the common substructure P via Theorem 3.3). So (S, D, C) is one slice of a structurally fertile axiom landscape, not an exhaustive axiomatisation. What distinguishes this particular slice is that it is structurally selectable: at $N = 5$ the canonical S+D+C witness (Remark 9, Appendix A) is fully determined by the topos-theoretic indicator reading of D plus the requirement that the magma’s automorphism be internalised in its operation. The other ten probe axioms have no such joint structural characterisation.

Categorical connections and their limits. The core $\text{core}(S)$ is a retract of the carrier via (s, r) in the standard sense of [3]; the three-category decomposition defines a functor from E2PM to three-block set partitions (Theorems 4.2, 4.4); D is a codomain-purity condition on the row maps $\{L_a^{\text{core}}\}$. The categorical analogy stops at composition: the row maps are *not* closed under composition (the no-associativity theorem predicts this – associativity would be a prerequisite for the row family to form a monoid), so standard categorical constructions (internal hom, enrichment, Turing morphism) cannot be stated internally. The composition-closure failure is a consequence of the same obstruction that motivates the setting.

Scope and limitations. Whether the three-capability separation appears in other settings (λ -calculus, typed systems, infinite algebras) is the primary open question. Theorem 3.10 constrains the target: any such translation must work without associativity or identity. Each capability admits multiple axiom forms; the independence result holds for all variants. The independence also implies

that any system combining all three capabilities must introduce additional axioms beyond S, D, and C to connect them; the nature of these connecting axioms is a natural direction for further work.

10 Conclusion

Four obstructions — K-infinity, no-associativity, asymmetry, and single-absorber collapse — jointly narrow the candidate class of finite magmas that can carry the operational capabilities of splitting, dichotomy, and composition down to the setting of this paper: finite, non-associative, non-commutative, extensional 2-pointed magmas. Within it, we have identified three natural algebraic properties, internal splitting (S), the classifier dichotomy (D), and the Internal Composition Property (C), and proved they are pairwise independent. The six non-implications are machine-checked in Lean 4. Joint irredundance strengthens this: all eight Boolean combinations of (S, D, C) are realized (Theorem 6.2), so no capability is a Boolean function of the other two. The optimal coexistence witness has $N=5$ (Theorem 7.2). The three-category decomposition induced by D is a functorial invariant (Theorem 4.2); most principal witnesses are role-rigid, with the canonical $N=5$ S+D+C witness as the deliberate exception (Proposition 4.6, Remark 9); at $N=5$ a structure theorem pins retraction, classifier, and C-triple onto the three core slots and rules out strong S altogether (Theorem 4.8), the mirror-row theorem (Theorem 4.11) bounds $|\text{Aut}(M)| \leq 2$, while at $N \geq 6$ strong S returns and rigidity fails (Proposition 4.12); and the ICP is logically equivalent to Compose+Inert (Theorem 5.1). Lean-formalized results — independence counterexamples, joint-irredundance witnesses, decomposition invariance, the ICP equivalence, the all- N scaling witness (Theorem 7.3), H-triple type-coherence (Theorem 3.12), the $N=5$ structure theorem (Theorem 4.8), and the mirror-row theorem (Theorem 4.11) — compile with zero sorry. The remaining universal- N algebraic results of Section 3 not yet transcribed to Lean (most notably the P-witness identity of Theorem 3.2 and the classifier-count bound of Theorem 3.13) are proved by hand and supported by Z3 UNSAT certificates.

Open problems.

- (1) *Cross-formalism universality*: Does the three-capability separation appear in the λ -calculus, typed settings, or infinite algebras? The no-associativity obstruction (Theorem 3.10) constrains the target setting.
- (2) *Categorical characterization of C*: The ICP is the weakest non-degenerate factorization property of the left-regular representation (weakening either pairwise-distinctness or non-triviality makes it vacuous). Is it equivalent to a standard universal property (e.g., existence of a partial internal hom)?
- (3) *Categorical characterization of D*: D is the unique coarsest non-trivial isomorphism-invariant partition of core (Corollary 4.3), isolating the two-valued aspect of a subobject classifier without the universality aspect. The left-multiplication maps lack composition closure, precluding standard categorical formulations. Does D correspond to a known concept in non-associative categorical algebra, or does it constitute a new structural property?
- (4) *Rigidity propagation beyond $N=5$* : Theorem 4.8 establishes a rigid role assignment at $N=5$ (and in particular rules out strong S), but rigidity fails at $N=6, 7, 8$ under strong S

(Proposition 4.12). Several natural single-axiom strengthenings (classifier-nontriviality on core, asymmetric anchoring of the section, anchor monotonicity, strict anchor asymmetry) were each tested and each Z3-satisfiable alongside a non-trivial automorphism at some $N \in \{6, 7, 8\}$ (scripts and witness JSONs under scripts/). Characterizing the minimum axiom system that uniformly forces role-rigidity at $N \geq 5$ — and whether it admits a bounded finite specification — is open.

Artifact. All Lean files, counterexample tables, and SAT reproduction scripts are available at <https://github.com/stefanopalmieri/finite-magma-independence>.

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A Cayley Tables

All tables are extracted from the Lean source files and verified by lake build. Element i 's row is row i ; entry (i, j) is $i \cdot j$.

$N=4$: *Minimal S+D witness* ($s = r$). Roles: $0 = z_1, 1 = z_2, 2 = \tau, 3 = s = r$. Lean: Dichotomic.lean, kripke4.

$\cdot$	0	1	2	3
0	0	0	0	0
1	1	1	1	1
2	0	1	0	1
3	0	0	2	3

$N=5$: *Minimal S+D witness* ($s \neq r$). Roles: $0 = z_1, 1 = z_2, 2 = s, 3 = r, 4 = \tau$. Lean: Dichotomic.lean, kripke5.

$\cdot$	0	1	2	3	4
0	0	0	0	0	0
1	1	1	1	1	1
2	1	0	3	4	2
3	0	2	4	2	3
4	0	1	1	0	0

$N=5$: *Canonical S+D+C Coexistence* ($s = r$, *minimal coexistence witness*). Roles: $0 = z_1$ (FALSE), $1 = z_2$ (TRUE), $2 = \tau_1$ (classifier, indicator of $\{z_1, z_2\}$, ICP a), $3 = \tau_2$ (classifier, indicator of $\{z_2, z_2\}$, ICP c), $4 = g = s = r$ (non-classifier acting on core as the transposition ($\tau_1 \tau_2$), ICP b). Lean: Witness5.lean.

·	0	1	2	3	4
0	0	0	0	0	0
1	1	1	1	1	1
2	0	1	1	0	0
3	0	1	0	1	0
4	0	1	3	2	4

$N=6$: *S+D+C Coexistence (role overlap)*. Roles: $0 = z_1$, $1 = z_2$, $2 = s$ (also ICP a), $3 = r$ (also ICP b), $4 = \tau$, $5 = \text{ICP } c$. Lean: Witness6.lean.

·	0	1	2	3	4	5
0	0	0	0	0	0	0
1	1	1	1	1	1	1
2	3	3	4	2	5	3
3	0	1	3	5	2	4
4	0	0	1	0	1	1
5	2	2	5	4	3	2

$N=8$: *S $\Rightarrow$ D Counterexample*. Roles: $0 = z_1$, $1 = z_2$, $2 = s$, $3 = r$, $4 = \text{classifier}$, $5, 7 = \text{mixed}$ (violate dichotomy). Lean: Countermodel1.lean.

·	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1
2	3	3	7	3	4	6	5	2
3	0	1	7	3	4	6	5	2
4	0	0	0	0	0	0	1	0
5	6	2	6	2	1	1	1	1
6	0	0	5	2	2	2	2	2
7	2	2	2	1	2	2	6	3

$N=6$: *S $\Rightarrow$ C Counterexample (structural)*. Roles: $0 = z_1$, $1 = z_2$, $2 = s = r$ (involution on core). Retraction pair holds; no triple satisfies ICP despite 4 core elements. Lean: E2PM.lean (sNoH6_e2pm).

·	0	1	2	3	4	5
0	0	0	0	0	0	0
1	1	1	1	1	1	1
2	0	3	3	2	5	4
3	2	4	5	5	1	4
4	5	3	0	0	3	2
5	4	2	2	2	2	2

$N=10$: *D $\Rightarrow$ C Counterexample*. Roles: $0 = z_1$, $1 = z_2$, $2 = s$, $3 = r$, $4 = \tau$. No triple satisfies ICP. Lean: Countermodels10.lean (dNotH), ICP.lean (dNotH_no_icp).

·	0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1	1
2	3	3	2	3	4	5	6	7	9	8
3	0	1	2	3	4	5	6	7	9	8
4	0	0	1	1	1	1	1	1	1	1
5	1	0	0	0	0	0	0	0	0	0
6	2	3	9	9	9	9	9	9	9	8
7	3	2	9	9	9	9	9	9	9	8
8	1	0	1	0	1	1	1	1	0	0
9	0	1	0	1	0	1	1	0	1	1

$N=10$: *C $\Rightarrow$ D Counterexample*. Roles: $0 = z_1$, $1 = z_2$, $2 = s$, $3 = r$, $4 = \tau$. ICP holds ($a=8, b=6, c=7$). Element 5 violates dichotomy. Lean: Countermodels10.lean (hNotD).

·	0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1	1
2	3	1	3	4	9	6	8	5	7	2
3	0	1	9	2	3	7	5	8	6	4
4	0	0	1	1	1	1	1	1	0	0
5	0	0	2	0	0	0	0	0	3	1
6	2	2	2	8	3	9	4	7	9	7
7	8	3	2	8	3	9	4	7	3	1
8	9	2	2	3	8	1	3	7	1	7
9	2	2	2	2	4	7	6	7	2	0

$N=4$: *D $\Rightarrow$ S Counterexample (tight)*. Element 2 is a classifier: row on core is $(1, 1) \subseteq \{z_1, z_2\}$. Element 3 is a non-classifier: row on core is $(2, 2) \subseteq \text{core}$. No retraction pair exists among the 4 candidate (s, r) assignments. Tight: D requires $N \geq 4$. Lean: E2PM.lean (d_not_implies_s_tight).

·	0	1	2	3
0	0	0	0	0
1	1	1	1	1
2	0	1	1	1
3	2	3	2	2

$N=5$: *C $\Rightarrow$ S Counterexample (tight)*. ICP holds but no retraction pair exists. Tight: ICP requires $N \geq 5$. Lean: E2PM.lean (h_not_implies_s_tight).

·	0	1	2	3	4
0	0	0	0	0	0
1	1	1	1	1	1
2	3	1	0	3	1
3	2	4	3	4	2
4	2	2	1	0	3

$N=5$: *C $\Rightarrow$ D Counterexample (tight)*. ICP holds but the dichotomy fails: no element maps core entirely to $\{z_1, z_2\}$ (no classifier exists). Tight: ICP requires $N \geq 5$. Lean: E2PM.lean (h_not_implies_d_tight).

·	0	1	2	3	4
0	0	0	0	0	0
1	1	1	1	1	1
2	3	3	4	3	3
3	2	4	4	4	3
4	2	2	2	4	4

$N=10$: *S+D+C Coexistence Witness (all roles distinct)*. S+D+C roles: $0 = z_1$, $1 = z_2$, $2 = s$, $3 = r$, $4 = \tau$, $8 = \eta$ (ICP a), $6 = g$ (ICP b , core-preserving), $7 = \rho$ (ICP c). Lean: Witness10.lean.

·	0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1	1
2	3	3	4	3	7	5	9	6	8	2
3	0	1	9	3	2	5	7	4	8	6
4	0	0	1	1	1	0	0	0	1	1
5	2	2	7	2	8	9	4	3	4	2
6	0	0	6	4	8	7	3	3	4	9
7	2	2	6	4	8	9	4	3	4	9
8	2	2	4	8	4	3	4	4	8	9
9	3	4	7	3	9	2	2	9	2	3

B Proof Inventory

All Lean files compile with `lake build`, zero sorry. Proof styles: *Algebraic* = pure equational reasoning (no decide); *decide* = verified

by kernel computation ($N \leq 8$); *native_decide* = verified by compiled native code ($N = 10$).

File	Thms	Style	Content
Dichotomic	20	Algebraic	Decomposition, bounds, classifier dichotomy
NoCommutativity	3	Algebraic	Asymmetry
Functoriality	4	Algebraic	Decomposition invariance
CapabilityInvariance	4	Algebraic	S, D, C each invariant
ICP	20	Alg.+decide	ICP $\Leftrightarrow$ Compose+Inert
Countermodel	5	decide	$S \not\Rightarrow D$ ($N=8$)
Countermodels10	9	native_dec.	$D \not\Rightarrow C$, $C \not\Rightarrow D$
E2PM	19	decide	$D \not\Rightarrow S$, $C \not\Rightarrow S$, $C \not\Rightarrow D$, $S \not\Rightarrow C$ (tight)
Witness10	6	native_dec.	S+D+C at $N=10$
Witness5	3	decide	S+D+C at $N=5$, no ICP at $N=4$
Witness6	3	decide	S+D+C at $N=6$
BooleanCube	25	decide+nat.	Joint irredundance: all 8 cells of (S, D, C)
Total	121		